Nicolás E. Díaz Ferreyra (Ph.D.)

Personal Information

Date of Birth and Nationality:	2 October 1987 | Argentine, German
Address:	Güntherße 9c, 22087, Hamburg, Germany
Phone:	+49 (0) 171 8325359
Website:	https://www.ndiaz-ferreyra.com/
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Research Areas:	privacy engineering, usable security, human-computer interaction, mining software repositories, empirical software engineering

Education

Feb. 2019 – Nov. 2015
Ph.D. in Computer Science | **Universität Duisburg-Essen**, Duisburg, Germany
Dissertation: "Instructional Awareness: A User-Centred Approach for Risk Communication in Social Network Sites" | Supervisors: Prof. Maritta Heisel and Prof. Nicole Krämer
Grade: 1.0 (Magna Cum Laude)

Dec. 2013 – March 2007
Msc. Degree in Information Systems and Engineering | **Universidad Tecnologica Nacional**, Argentina
Thesis: "Development of a Tool for the Analysis of Variability in Software Product Lines" | Advisors: Prof. Silvio Gonnet and Prof. Horacio Leone
Grade: 10 (Outstanding)

Research Experience

Current – Nov. 2021
SENIOR RESEARCHER AND LECTURER (Oberingenieur) | **Hamburg University of Technology**, Hamburg, Germany
Institute of Software Security

- Associate researcher in "Sec4AI4Sec: Cybersecurity for AI-Augmented Systems", EU H2020 101120393
- Associate researcher in "AssureMOSS: Assurance and Certification in Secure Software", EU H2020 952647

Oct. 2021 – March 2019
POST-DOCTORAL RESEARCHER | **Universität Duisburg-Essen**, Duisburg, Germany
Department of Computer Science and Applied Cognitive Science

- Coordinator of the Research Training Group "User-Centred Social Media", DFG GRK 2167
- Associate researcher in "PDP4E: Privacy and Data Protection for Engineering", EU H2020 787034

Feb. 2019 – Nov. 2015
RESEARCH FELLOW (Ph.D candidate) | **Universität Duisburg-Essen**, Duisburg, Germany
Department of Computer Science and Applied Cognitive Science

July 2015 – Feb. 2014
ASSOCIATE SOFTWARE DEVELOPER | **Milestone Systems**, Copenhagen, Denmark
Incubation Business Unit

Academic Services, Teaching and Talks

CURRENT TEACHING DUTIES

- SEMINARS: "Usable Security and Privacy" (TUHH, SoSe); "Engineering Privacy-Friendly Systems" (TUHH, WiSe)
- MSC. COURSES: "Secure Software Engineering" (TUHH, SoSe);"Cybersecurity Data Science" (TUHH, SoSe); "Software Security" (TUHH, WiSe).
- BSC. COURSES: "Introduction to Information Security" (TUHH, WiSe).

RECENT AND UPCOMING PRESENTATIONS AND TALKS

- Keynote speaker at the 20th IFIP Summer School on Privacy and Identity Management | Copenhagen, Denmark (September 2025)
- Invited talk at the University of Paderborn: "Supporting Privacy in Networked Environments: Lessons from the Facebook Saga and Open Challenges for the GitHub Sequel" | Paderborn, Germany (July 2024)
- Technical speaker at the International Conference on Technical Debt: Emerging Researchers' Forum | Lisbon, Portugal (May 2024)
- Invited talks at the University of Melbourne, RMIT University, Monash University | Melbourne, Australia (Sep. - Oct. 2023)
- Invited talk at the The University of New South Wales | Canberra, Australia (Oct. 2023)

PROGRAM COMMITTEE AND PEER REVIEWING SERVICES

- CONFERENCES: CHASE 2026-2025, EASE 2026-2024, SANER 2026, ICSE-SRC 2027-2026, PST 2026, SCP 2026, EuroUSEC 2025-2022, ARES 2026-2023, IEEE CSR 2026-2021, IFIP SEC 2022-2020.
- WORKSHOPS: Fairness 2026 (colocated with SANER), IWPE 2026-2022 (co-located with IEEE EuroS&P), DPM 2023-2021 (co-located with ESORICS), SVM 2026-2023 (co-located with ICSE).
- JOURNALS: IEEE TSE, IEEE Software, JSS, EMSE, IST, COSE, SoSyM, AUSE, IJHCI, CHBR, SNAM.

OTHER ACADEMIC AND PROFESSIONAL SERVICES

- Proceedings Co-Chair at the Int. Conference on Cooperative and Human Aspects of Software Engineering (CHASE '26)
- Jopurnal First Track Chair at the European Conference on Software Architecture (ECSA '26)

- Student Research Competition Co-Chair at the Int. Conference on the Foundations of Software Engineering (FSE '26)
- Publicity chair of the 8th International Conference on Technical Debt (TechDebt '25)
- Co-organizer of "MSR4P&S: Int. Workshop on MSR Applications for Privacy and Security" at FSE '22, SANER '24 '25 and '26.

Acquired Funding and Project Grants

April 2026	Project "DevSSATD: Developer-Centred Management of Self-Admitted Technical Debt for Security" | 3-year DFG Individual Research Grant (€360,000)
May. 2025	Project "KiThreat: Threat and Risk Assessment Methods for GenAI-Based Software" | 6-month Contract Research Project (Industry-Funded, €12,000)

Training & Development

April. 2023	Course on Structural Equation Modeling in HCI Research using SEMinR | Hamburg, Germany
June. 2022	International Summer School on Search- and Machine Learning-based Software Engineering | Cordoba, Spain
Sep. 2019	15th Reasoning Web Summer School on Explainable AI | Bolzano, Italy
May. 2018	10th South School on Internet Governance at the Organization of the American States | Washington DC, U.S.A
Dec. 2017	2nd Winter School in Engineering and Computer Science on Formal Verification | Jerusalem, Israel
July 2017	Dagstuhl Seminar No. 17301 on "User-Generated Content in Social Media" | Wadern, Germany
June 2016	1st Interdisciplinary Summer School on Privacy | Berg en Dal, The Netherlands

Scholarships and Awards

Sep. 2023	MSCA-RISE OpenInnoTrain secondment at RMIT University, Australia | *Academic exchange*
May. 2022	Best Data and Tool Showcase Paper Award at the 16th International Conference on Mining Software Repositories.
Sep. 2019	Stiftung MERCATOR: "Global Young Faculty" support program for young researchers | *Scholarship*
Oct. 2016	DAAD German-Tunisian "Change by Exchange" cooperation program | *Academic exchange*

Languages

Spanish:	Mothertongue
English:	Level "C1" (Proficient User) in understanding (listening-reading), speaking, and writing
German:	Level "B1" (Intermediate User) in understanding (listening-reading), speaking, and writing

Computer Science Skills

Formal Modelling:	OCL, Z, Petri nets, UML, OVM, BPMN, Problem Frames
Programming:	C, C++, C#, Java, R, JavaScript, SQL, Python, MatLab, Prolog, Lingo, GPSS, Vensim, SHELL Scripting, JSON, HTML/CSS, Android

Selected Publications

[1] N. E. Díaz Ferreyra, M. S. Gurupathi, Z. Codabux, N. Arachchilage, and R. Scandariato, "Security Concerns in Generative AI Coding Assistants: Insights from Online Discussions on GitHub Copilot," in *Proceedings of the 2026 30th International Conference on Evaluation and Assessment in Software Engineering Companion*, 2026.

[2] C. Tony, N. E. Díaz Ferreyra, M. Mutas, S. Dhiff, and R. Scandariato, "Prompting Techniques for Secure Code Generation: A Systematic Investigation," *ACM Transactions on Software Engineering and Methodology*, 2025. doi: 10.1145/3722108.

[3] N. E. Díaz Ferreyra, S. Khelifi, N. Arachchilage, and R. Scandariato, "The Good, the Bad, and the (Un)Usable: A Rapid Literature Review on Privacy as Code," in *18th International Conference on Cooperative and Human Aspects of Software Engineering (CHASE 2025)*, 2025.

[4] M. Mock, J. Melegati, M. Kretschmann, N. E. Diaz Ferreyra, and B. Russo, "MADE-WIC: Multiple Annotated Datasets for Exploring Weaknesses In Code," in *Proceedings of the 39th IEEE/ACM International Conference on Automated Software Engineering*, Sacramento, CA, USA: Association for Computing Machinery, 2024, pp. 2346–2349, isbn: 9798400712487. doi: 10.1145/3691620.3695348.

[5] N. E. Díaz Ferreyra, M. Shahin, M. Zahedi, S. Quadri, and R. Scandariato, "What Can Self-Admitted Technical Debt Tell Us About Security? A Mixed-Methods Study," in *Proceedings of the 21st International Conference on Mining Software Repositories (MSR '24)*, 2024.

[6] C. Tony, M. Balasubramanian, N. E. Díaz Ferreyra, and R. Scandariato, "Conversational DevBots for Secure Programming: An Empirical Study on SKF Chatbot," in *Evaluation and Assessment in Software Engineering (EASE 2022)*, ACM, 2022.

[7] C. Tony, N. Díaz Ferreyra, and R. Scandariato, "GitHub Considered Harmful? Analyzing Open-Source Projects for the Automatic Generation of Cryptographic API Call Sequences," in *2022 IEEE 22nd International Conference on Software Quality, Reliability and Security Companion (QRS-C)*, 2022.

[8] P. Billawa, A. Bambhore Tukaram, N. E. Díaz Ferreyra, J.-P. Steghöfer, G. Simhandl, and R. Scandariato, "SoK: Security of Microservice Applications: A Practitioners' Perspective on Challenges and Best Practices," in *Proceedings of ARES '22*, 2022.

[9] Q.-C. Bui, R. Scandariato, and N. E. Díaz Ferreyra, "Vul4J: A Dataset of Reproducible Java Vulnerabilities Geared Towards the Study of Program Repair Techniques," in *International Conference on Mining Software Repositories (MSR '22)*, 2022.

[10] N. E. Díaz Ferreyra, M. Vidoni, M. Heisel, and R. Scandariato, "Cybersecurity Discussions in Stack Overflow: A Developer-Centred Analysis of Engagement and Self-Disclosure Behaviour," *Social Network Analysis and Mining*, vol. 14, no. 1, p. 16, Dec. 2023.

[11] C. Tony, M. Mutas, N. E. Díaz Ferreyra, and R. Scandariato, "LLMSecEval: A Dataset of Natural Language Prompts for Security Evaluations," in *Proceedings of the 20th International Conference on Mining Software Repositories (MSR '23)*, 2023.

[12] K. Russel Go, S. Soundarapandian, A. Mitra, M. Vidoni, and N. E. Díaz Ferreyra, "Simple Stupid Insecure Practices and GitHub's Code Search: A Looming Threat?" *Journal of Systems and Software*, p. 111 698, 2023.

[13] N. E. Díaz Ferreyra, A. Imine, M. Vidoni, and R. Scandariato, "Developers Need Protection, Too: Perspectives and Research Challenges for Privacy in Social Coding Platforms," in *16th Int. Conf. on Cooperative and Human Aspects of Software Engineering (CHASE '23)*, 2023.